

STA 332/MATH 343: Homework 4

Duke University, Fall 2026

Due: October 3rd, 11:59 PM on Gradescope

UMVUE, Completeness, and Exponential Families

1. Review: give brief but formal definitions of the following concepts for a statistical model $\{f_\theta : \theta \in \Theta\}$:
 - (a) Ancillary statistic
 - (b) Sufficient statistic
 - (c) Point estimator of $g(\theta)$;
 - (d) Unbiased estimator of $g(\theta)$;
 - (e) Uniformly minimum-variance unbiased estimator of $g(\theta)$
2. Properties of conditional expectations
 - (a) Let U be a discrete random variable. Show that $\mathbb{E}[f(U) \mid U] = f(U)$.
 - (b) Let X and Y be independent continuous random variables. Show that $\mathbb{E}[X \mid Y] = \mathbb{E}[X]$.
 - (c) Let $X_1, X_2 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, 1)$, so that $U = (X_1 + X_2)/2$ is an unbiased estimator for θ . Let $S = X_2$ and compute $\mathbb{E}_\theta[U \mid S]$. Is it a statistic? What can we conclude about whether S is sufficient?
3. A factory inspects $n \geq 2$ independently produced components. Let $X_i = 1$ if component i passes inspection and $X_i = 0$ otherwise. Suppose $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p)$, where the pass probability $0 < p < 1$ is unknown.
 - (a) Show that $\text{Ber}(p)$ is an exponential family, and use this to show $T = \sum_{i=1}^n X_i$ is sufficient and complete.
 - (b) Show that $V = (T, X_1)$ is not complete by finding a nonzero function of V whose expectation is zero for every p .
 - (c) Find the UMVUE of p^2 by conditioning $X_1 X_2$ on T . Make sure to consider the cases $T < 2$ and $T \geq 2$ separately.
4. A workshop cuts n square panels with dimensions $X_1 \times X_1, X_2 \times X_2, \dots, X_n \times X_n$, where $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, \theta)$, and $\theta > 0$ is the largest permitted side length. We want to estimate θ^2 , the area of the largest permitted panel. Let

$$U = 3X_1^2 \quad \text{and} \quad M = X_{(n)}.$$

- (a) Show that U is unbiased for θ^2 .
- (b) Compute the UMVUE for θ^2 by conditioning U on M . You may use the following conditional description:

$$(X_1 \mid M = m) = \begin{cases} m, & \text{with prob. } \frac{1}{n}, \\ \text{Unif}(0, m), & \text{with prob. } 1 - \frac{1}{n}. \end{cases}$$

5. A maintenance team records independent repair durations $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, 1/\theta)$, where $n \geq 2$, the shape $\alpha > 0$ is known, and the scale $\theta > 0$ is unknown. Write $S = \sum_i X_i$ and $M = X_{(n)}$. Recall from Lecture 5 and HW2 that $X_i = \theta Z_i$ with $Z_i \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, 1)$, and

$$f_\theta(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)} \mathbb{1}_{\{x>0\}} \exp\left\{-\frac{x}{\theta} - \alpha \log \theta\right\}.$$

- Use this representation to establish that S is complete and sufficient, and hence minimal sufficient.
 - Show that (M, S) is not complete. *Hint: consider the function $a(M, S) = M/S$. Show its expectation is some c which does not depend on θ .*
 - Find the UMVUE of θ . You may use $S \sim \text{Gamma}(n\alpha, 1/\theta)$.
 - Now suppose $n\alpha > 2$. Compute $\mathbb{E}_\theta[S^{-1}]$ and $\mathbb{E}_\theta[S^{-2}]$, and find the UMVUE of the rate $1/\theta$. Is the reciprocal of the UMVUE from part (c) also unbiased for $1/\theta$?
6. A sensor gives independent signed readings $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \theta^2)$ of a signal, where $\theta > 0$ is the mean response and the standard deviation is also θ . Assume $n \geq 2$. Define $T_1 = \sum_i X_i$ and $T_2 = \sum_i X_i^2$.
- Show that $\mathcal{N}(\theta, \theta^2)$ is not a one-parameter exponential family.
 - Show that θ is a scale parameter and that (T_1, T_2) is minimal sufficient.
 - Compute $\mathbb{E}_\theta[T_1]$, $\mathbb{E}_\theta[T_1^2]$, and $\mathbb{E}_\theta[T_2]$.
 - Use your answers to construct a nonzero function of (T_1, T_2) with expectation zero for every θ . What does this say about completeness?
7. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(\alpha, \beta)$, with density

$$f_{\alpha, \beta}(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbb{1}_{\{0 < x < 1\}}.$$

- Suppose $\alpha > 0$ is known and $\beta > 0$ is unknown. Write the density as a one-parameter exponential family. Identify a complete sufficient statistic.
 - Now suppose $\beta > 0$ is known and $\alpha > 0$ is unknown. Repeat part (a).
8. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where μ is known and $\sigma^2 > 0$ is unknown. In HW3 you found the MLE $\widehat{\sigma^2} = n^{-1} \sum_i (X_i - \mu)^2$.
- Write this model as a one-parameter exponential family in σ^2 . Identify a complete sufficient statistic.
 - Show that the MLE above is also the UMVUE of σ^2 .
9. Let $X_1, \dots, X_n$ be i.i.d. with density

$$f_\theta(x) = \frac{1}{\theta} (1+x)^{-(1+\theta)/\theta} \mathbb{1}_{\{x>0\}}, \quad \theta > 0.$$

- Determine the distribution of $Y_i = \log(1 + X_i)$.
- Write the original density $f_\theta(x)$ as a one-parameter exponential family. Identify a complete sufficient statistic T for θ , and conclude that it is minimal sufficient.

- (c) Determine the distribution of T and find the UMVUE of θ .
- (d) Find the MLE of θ and compare it with the UMVUE you found.

10. Let $X_1, \dots, X_n$ be i.i.d. with density

$$f_{\theta}(x) = \frac{1}{x \log(\theta)} \mathbb{1}_{\{1 < x < \theta\}}, \quad \theta > 1,$$

and let $M = X_{(n)}$.

- (a) Prove that M is sufficient for θ .
- (b) Find the cdf and density of M .
- (c) Prove directly that M is complete, and use this to conclude that M is minimal sufficient.
Hint: if $\mathbb{E}_{\theta}[a(M)] = 0$ for every $\theta > 1$, multiply by $[\log \theta]^n$ and differentiate the resulting integral with respect to its upper limit.
- (d) Find the UMVUE of $\log \theta$. *Hint: show $\log X_i \sim \text{Unif}(0, \log \theta)$ and use the expectation calculation from Example 3, Lecture 10 without reproving it.*